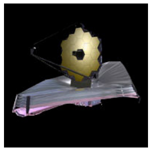

9493 - SN Eos: A Multiply-Imaged, 30x Magnified SN Near the Epoch of Reionization

Cycle: 4, Proposal Category: DD

INVESTIGATORS

Name	*Institution*
Dr. David Coulter (PI)	**Space Telescope Science Institute**
Conor Larison (CoI) (CoPI)	Space Telescope Science Institute
Dr. Joseph Frederick Victor Allingham (CoI) (CoPI)	Ben-Gurion University of the Negev
Dr. Vasily Kokorev (CoI) (CoPI)	University of Texas at Austin
Dr. Justin Pierel (CoI)	Space Telescope Science Institute
Massimo Pascale (CoI)	University of California - Los Angeles
Dr. Johan Pierre Richard (CoI) (ESA Member)	Centre de Recherche Astrophysique de Lyon
Prof. Masamune Oguri (CoI)	Chiba University
Dr. Lukas Jonathan Furtak (CoI)	University of Texas at Austin
Prof. Adi Zitrin (CoI)	Ben-Gurion University of the Negev
Tom Resseguier (CoI)	Space Telescope Science Institute
Miriam Golubchik (CoI)	Ben-Gurion University of the Negev
Dr. Eiichi Egami (CoI)	University of Arizona
Dr. Seiji Fujimoto (CoI) (CSA Member)	University of Toronto
Dr. Dan Coe (CoI) (ESA Member)	Space Telescope Science Institute - ESA - JWST
Dr. Abdurro'uf Abdurro'uf (CoI)	Indiana University System
Dr. Gabriel Brammer (CoI) (ESA Member)	University of Copenhagen, Niels Bohr Institute
Prof. Steven L. Finkelstein (CoI)	University of Texas at Austin
Dr. Anton M. Koekemoer (CoI)	Space Telescope Science Institute
Hollis Akins (CoI)	University of Texas at Austin
Ray A. Lucas (CoI)	Space Telescope Science Institute

Name	***Institution***
Christa Noel DeCoursey (CoI)	University of Arizona
Aadya Agrawal (CoI)	University of Illinois at Urbana - Champaign
Dr. Stefan Schuldt (CoI) (ESA Member)	Universita di Milano
Prof. Sherry H. Suyu (CoI) (ESA Member)	Technical University of Munich
Dr. Armin Rest (CoI)	Space Telescope Science Institute
Dr. Saurabh W. Jha (CoI)	Rutgers the State University of New Jersey
Dr. Jose M. Diego (CoI) (ESA Member)	Instituto de Fisica de Cantabria
Dr. Fengwu Sun (CoI)	Harvard University
Dr. Louis-Gregory Strolger (CoI)	Space Telescope Science Institute
Mike Engesser (CoI)	Space Telescope Science Institute
Prof. Liang Dai (CoI)	University of California - Berkeley
Dr. Larry Bradley (CoI)	Space Telescope Science Institute
Dr. James M DerKacy (CoI)	Space Telescope Science Institute
Dr. Tiger Hsiao (CoI)	University of Texas at Austin
Prof. John Chisholm (CoI)	University of Texas at Austin
Mr. Mateusz Bronikowski (CoI)	University of Nova Gorica
Dr. Tanja Petrushevska (CoI)	University of Nova Gorica
Thomas Moore (CoI)	Space Telescope Science Institute
Dr. Rohan Naidu (CoI)	Massachusetts Institute of Technology
Dr. Suhail Dhawan (CoI) (ESA Member)	University of Cambridge
Dr. Jorryt Matthee (CoI) (ESA Member)	Institute of Science and Technology Austria
Dr. Patrick Kelly (CoI)	University of Minnesota - Twin Cities
Dr. Franz Bauer (CoI)	Space Science Institute
Dr. Francesco Maria Valentino (CoI) (ESA Member)	Technical University of Denmark-DTU Space
Prof. Robert M. Quimby (CoI)	San Diego State University
Dr. Jeyhan Kartaltepe (CoI)	Rochester Institute of Technology
Dr. Lindsey Kwok (CoI)	Northwestern University
Dr. Massimo Griggio (CoI)	Space Telescope Science Institute
Dr. Melissa Shahbandeh (CoI)	Space Telescope Science Institute
Dr. Ori Dosovitz Fox (CoI)	Space Telescope Science Institute
Dr. Estefania Padilla Gonzalez (CoI)	The Johns Hopkins University
Dr. Matthew Ryan Siebert (CoI)	Space Telescope Science Institute

OBSERVATIONS

Folder	***Observation***	***Label***	***Observing Template***	***Science Target***
Lensed SN NIRCam				
	1	NIRCam	NIRCam Imaging	(1) SN_Eos_Image_101.1
	2	NIRSpec	NIRSpec Fixed Slit Spectroscopy	(1) SN_Eos_Image_101.1
	3	NIRSpec	NIRSpec Fixed Slit Spectroscopy	(2) SN_Eos_Image_101.2

ABSTRACT

We have discovered a z_spec = 5.13, multiply-imaged supernova (SN) candidate (~24.8 AB F444W mag) in JWST NIRcam imaging for the Vast Exploration for Nascent, Unexplored Sources program (VENUS, PID 6882) on September 1. We dub this candidate SN Eos – named after the Titan goddess of dawn. Archival HST imaging, spanning 13 years to the current JWST dataset, rules out all static background sources (e.g., AGN or LRDs), and the identical SEDs of each image (expected as their relative time delay is ~rest-frame hours) rules out foreground sources (e.g., binary brown dwarfs). The current data point to a type IIP SN currently in the “plateau” phase of its light curve evolution in a low-metallicity environment, and our observations will offer a true test of whether local, low-metallicity SNe IIP are indeed true high-z analogues. The spectroscopic redshift and high combined magnification of the detected images (~60 when combining images) means that this is the first z > 5 SN ever discovered for which we can obtain a high signal-to-noise (S/N > 20) spectrum at rest-frame optical wavelengths, and includes a chance to make a direct measurement of metallicity near the Epoch of Reionization. Given its highly redshifted SED, only JWST can classify and characterize SN Eos, and observations are needed *now* to get the S/N required before SN Eos declines. This 10.1 hour program provides a crucial step toward understanding the first stars and fulfilling one of JWST's core mission objectives.

OBSERVING DESCRIPTION

NIRCam: We request 3 filter pairs to provide precise calibration for NIRSpec while helping to determine the light curve phase (long-wavelength), and to capture the diffuse Lya emission of the host (short-wavelength). We use F277W, F356W, F444W to match existing VENUS observations of SN Eos with minimum S/N ~20 (~800s per filter pair), and in the short-wavelength F070W (x3) as the only JWST filter overlapping with Lya. As described in the proposal text, Lya could be detectable in F070W, and a detection would show the spatial extent of the host without SN contamination. The exposure time requirement comes from our desired photometric precision in the long-wavelength filters (S/N ~20) given the potential for a ~1 mag decline in the light curve.

NIRSpec: Our program obtains a NIRSpec/Prism (fixed-slit) spectrum of each visible SN image with a minimum requirement of S/N =15-20 at

~3um so that we may accurately characterize any present Fe II 5018 AA line at ~3-sigma in each spectrum. Obtaining both spectra will unequivocally show SN Eos is multiply-imaged, and enable analysis of any spectral evolution from the short relative time delay. Then, as the time delay is of order rest-frame ~hours, we will combine the two spectra for analysis of long-lived spectral lines, in particular a ~10-sigma detection of Fe II if present. We opt to prioritize S/N in the spectrum while maximizing resolution with sub-pixel dithers, and find this is the optimal choice over medium resolution and/or the IFU.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	SN_Eos_Image_101.1	RA: 19 31 49.2176 (292.9550733d) Dec: -26 34 29.46 (-26.57485d) Equinox: J2000		
Comments: Category=Star Description=[Supernovae]				
(2)	SN_Eos_Image_101.2	RA: 19 31 49.1820 (292.9549250d) Dec: -26 34 28.65 (-26.57463d) Equinox: J2000		
Comments: Category=Star Description=[Supernovae]				
(3)	TA_Star	RA: 19 31 48.6498 (292.9527075d) Dec: -26 34 29.26 (-26.57479d) Equinox: J2000		
Comments: Category=Star Description=[O stars]				
(4)	MACS1931	RA: 19 31 49.6000 (292.9566667d) Dec: -26 34 34.00 (-26.57611d) Equinox: J2000		
Comments: Category=Clusters of Galaxies Description=[Rich clusters]				
(5)	SN_Eos_Image_101.3	RA: 19 31 48.3138 (292.9513075d) Dec: -26 34 40.07 (-26.57780d) Equinox: J2000		
Comments: Category=Star Description=[Supernovae]				
(6)	SN_Eos_Image_101.4	RA: 19 31 48.9477 (292.9539487d) Dec: -26 34 45.94 (-26.57943d) Equinox: J2000		
Comments: Category=Star Description=[Supernovae]				
(7)	SN_Eos_Image_101.5	RA: 19 31 52.4378 (292.9684908d) Dec: -26 34 32.26 (-26.57563d) Equinox: J2000		
Comments: Category=Star Description=[Supernovae]				

Observation

Proposal 9493, Observation 1: NIRCam

Thu Nov 06 20:00:20 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	SN_Eos_Image_101.1	RA: 19 31 49.2176 (292.9550733d) Dec: -26 34 29.46 (-26.57485d) Equinox: J2000		

Comments:
Category=Star
Description=[Supernovae]

Template

Module	Subarray
B	FULL

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	INTRAMODULEX	3	STANDARD		1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F070W	F277W	BRIGHT2	6	2	6	3	805.258	
2	F070W	F356W	BRIGHT2	6	2	6	3	805.258	
3	F070W	F444W	BRIGHT2	6	2	6	3	805.258	

Special Requirements

Before Date 01-NOV-2025:00:00:00
Offset 45.0 arcsec, 40.0 arcsec

Group Observations 1, 2, 3, Non-interruptible

Observation

Proposal 9493, Observation 2: NIRSpec

Thu Nov 06 20:00:20 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(NIRSpec (Obs 2)) Warning (Form): Observers are responsible for checking that target acquisition is feasible.

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	SN_Eos_Image_101.1	RA: 19 31 49.2176 (292.9550733d) Dec: -26 34 29.46 (-26.57485d) Equinox: J2000		

Comments:
Category=Star
Description=[Supernovae]

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	3 TA_Star	WATA	SUB2048	F110W	NRSRAPID	3	1	1	3.628	

Template

HFF Readout Mode	Slit	Subarray
false	S200A1	FULL

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	5	BOTH

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	S200A1	NRSIRS2	7	1	1	NONE	20	20	10504.001	

Special Requirements	Group Observations 1, 2, 3, Non-interruptible

Observation

Proposal 9493, Observation 3: NIRSpec

Thu Nov 06 20:00:20 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(NIRSpec (Obs 3)) Warning (Form): Observers are responsible for checking that target acquisition is feasible.

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	SN_Eos_Image_101.2	RA: 19 31 49.1820 (292.9549250d) Dec: -26 34 28.65 (-26.57463d) Equinox: J2000		

Comments:
Category=Star
Description=[Supernovae]

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	3 TA_Star	WATA	SUB2048	F110W	NRSRAPID	3	1	1	3.628	

Template

HFF Readout Mode	Slit	Subarray
false	S200A1	FULL

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	5	BOTH

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	S200A1	NRSIRS2	7	1	1	NONE	20	20	10504.001	

Special Requirements	Group Observations 1, 2, 3, Non-interruptible